

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO2: Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2,BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:100

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I _____ (Name / Reg No) ~~certify that this submission is my original~~
work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary
action.

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

Q1. Selection Sort – Analytical Problem

Apply the Brute Force technique using Selection Sort on the dataset [64, 25, 12, 22, 11]. Clearly interpret the problem and justify why Selection Sort is suitable. Perform the sorting step-by-step and show intermediate results after each pass. Count the total number of comparisons made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output. Interpret the efficiency of the algorithm based on the results.

Q2. Closest Pair – Brute Force Analysis

Given the set of points $\{(2,3), (12,30), (40,50), (5,1), (12,10), (3,4)\}$, apply the Closest Pair Brute Force algorithm. Interpret the problem and justify the method selection. Compute distances between all pairs and show the calculations clearly. Identify the closest pair and count the number of comparisons. Analyze the time complexity of the approach. Verify the correctness of the result. Interpret its applicability in real-world scenarios.

Q3. Convex Hull – Brute Force Approach

Apply the Brute Force Convex Hull algorithm on the set $\{(0,3), (2,2), (1,1), (2,1), (3,0), (0,0), (3,3)\}$. Interpret the problem and explain the concept of convex hull. Check all possible line pairs and identify valid hull edges. Show the logical steps used to determine the boundary. Count the total comparisons performed. Analyze the time complexity of the method. Interpret the result and its practical significance.

Q4. Searching Techniques – Comparative Analysis

Analyze the performance of Linear Search on unsorted data and Binary Search on sorted data. Interpret the problem and explain the working of both algorithms. Apply both methods and count the number of comparisons required. Analyze their time complexities in different cases. Compare their efficiency based on dataset conditions. Justify when each method is preferable. Interpret the results in terms of real-world applications.

Q5. Closest Pair (Real-Time Scenario)

A set of GPS coordinates represents emergency center locations. Apply the Closest Pair Brute Force algorithm to identify the nearest centers. Interpret the problem and justify the chosen method. Compute distances between all pairs and identify the minimum. Count the total comparisons made. Analyze the time complexity of the approach. Verify the correctness of the result. Interpret its importance in real-time applications.

Q6. Bubble Sort – Analytical Problem

Apply the Brute Force technique using Bubble Sort on the dataset [45, 12, 78, 34, 23]. Clearly interpret the problem and justify why Bubble Sort is used. Perform the sorting step-by-step and show intermediate results after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output. Interpret the efficiency of the algorithm.

Q7. Linear Search – Analytical Problem

Given an unsorted dataset [23, 45, 12, 67, 34, 89, 10], apply Linear Search to find the element 34. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to locate the element. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the result. Interpret the efficiency of Linear Search in real-world scenarios.

Q8. String Matching – Brute Force Analysis

Given a text "ABABABCABABABCAB" and a pattern "ABABC", apply the Brute Force string matching algorithm. Clearly interpret the problem and justify the method selection. Compare the pattern with the text at each position and show all matching steps. Count the number of comparisons performed. Identify all occurrences of the pattern. Analyze the time complexity of the approach. Interpret its efficiency compared to optimized methods.

Q9. Brute Force Maximum Subarray Problem

Given an array [-2, 3, 4, -1, -2, 1, 5, -3], apply the brute force approach to find the maximum subarray sum. Clearly interpret the problem and explain the method used to examine all subarrays. Show the calculations for different subarrays. Count the number of comparisons made. Identify the maximum sum and verify the result. Analyze the time complexity. Interpret the efficiency of this approach.

Q10. Nearest Neighbor Distance (2D Points)

Given a set of points $\{(1,2), (4,6), (7,8), (2,3), (5,5)\}$, apply the brute force approach to compute the distance between all pairs. Clearly interpret the problem and justify the method. Show all distance calculations step-by-step. Identify the nearest pair and count the number of comparisons. Analyze the time complexity of the approach. Verify the correctness of the result. Interpret its significance in spatial data analysis.

Q11. Selection Sort – Mixed Positive and Negative Numbers

Apply the Brute Force technique using Selection Sort on the dataset $[18, -4, 25, 0, -9, 13]$. Clearly interpret the problem and justify why Selection Sort is suitable for this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q12. Bubble Sort – Duplicate Values Analysis

Apply the Brute Force technique using Bubble Sort on the dataset $[32, 14, 32, 7, 19, 7]$. Clearly interpret the problem and explain why Bubble Sort can be applied to datasets with duplicate values. Perform the sorting process step-by-step and display intermediate results after each pass. Count the number of comparisons and swaps performed. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on repeated elements.

Q13. Linear Search – Element at Last Position

Given the unsorted dataset $[41, 26, 13, 57, 88, 34, 72]$, apply Linear Search to find the element 72. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required until the element is found. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears at the last position.

Q14. Binary Search – Middle Element Analysis

Given the sorted dataset $[6, 12, 18, 24, 30, 36, 42]$, apply Binary Search to find the element 24. Clearly interpret the problem and justify why Binary Search is suitable for this dataset. Show the step-by-step division of the search interval. Count the number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search when the target appears near the middle.

Q15. Closest Pair – Simple Coordinate Set

Given the set of points $\{(1,2), (4,6), (5,5), (9,1), (2,3)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the method selection. Compute the distance between every possible pair and show the calculations clearly. Identify the closest pair and count the total number of comparisons.

Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its usefulness in geometric applications.

Q16. Convex Hull – Basic Polygon Boundary Detection

Apply the Brute Force Convex Hull algorithm on the set $\{(0,0), (2,1), (4,0), (3,3), (1,4), (2,2)\}$. Clearly interpret the problem and explain the concept of convex hull formation. Check all possible line pairs and identify which edges belong to the hull. Show the logical reasoning used to determine the boundary points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q17. Selection Sort – Nearly Sorted Dataset

Apply the Brute Force technique using Selection Sort on the dataset [5, 9, 11, 14, 8, 20]. Clearly interpret the problem and justify the use of Selection Sort for this dataset. Perform the sorting step-by-step and show intermediate results after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on a nearly sorted list.

Q18. Bubble Sort – Reverse Ordered List

Apply the Brute Force technique using Bubble Sort on the dataset [90, 75, 60, 45, 30, 15]. Clearly interpret the problem and justify why Bubble Sort is appropriate for stepwise comparison-based sorting. Perform the sorting process pass by pass and display the intermediate arrays. Count the number of comparisons and swaps required. Analyze the worst-case time complexity. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm for reverse ordered data.

Q19. Linear Search – Element Not Present

Given the unsorted dataset [17, 29, 43, 56, 68, 71], apply Linear Search to find the element 50. Clearly interpret the problem and explain the search process step-by-step. Count the total number of comparisons made before concluding that the element is absent. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the search is unsuccessful.

Q20. Binary Search – First Element Detection

Given the sorted dataset [3, 8, 14, 19, 25, 31, 40], apply Binary Search to find the element 3. Clearly interpret the problem and justify why Binary Search is suitable. Show each step involved in narrowing the search interval. Count the number of comparisons required to find the target. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm when the element is at the beginning.

Q21. Closest Pair – Negative and Positive Coordinates

Given the set of points $\{(-2,-1), (3,4), (0,0), (5,1), (-1,-2)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the chosen method. Compute distances between all possible pairs and present the calculations clearly. Identify the closest pair and count the number of pairwise comparisons. Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its relevance in spatial analysis.

Q22. Convex Hull – Points with Interior Elements

Apply the Brute Force Convex Hull algorithm on the set $\{(1,1), (2,5), (5,4), (6,1), (3,3), (4,2)\}$. Clearly interpret the problem and explain the concept of outer boundary detection. Check all line pairs and determine which points form the convex hull while excluding interior points. Show the reasoning used to validate hull edges. Count the total comparisons performed. Analyze the time complexity and interpret the result in computational geometry.

Q23. Selection Sort – Repeated Minimum Elements

Apply the Brute Force technique using Selection Sort on the dataset [12, 5, 7, 5, 18, 9]. Clearly interpret the problem and explain how Selection Sort handles repeated minimum values. Perform the sorting step-by-step and show the intermediate array after each pass. Count the number of comparisons and swaps made. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the algorithm's behavior on duplicated minima.

Q24. Bubble Sort – Already Sorted Dataset

Apply the Brute Force technique using Bubble Sort on the dataset [4, 8, 12, 16, 20, 24]. Clearly interpret the problem and justify why this case is important for algorithm analysis. Perform the sorting pass by pass and show the intermediate results. Count the number of comparisons and swaps made during execution. Analyze the best-case time complexity. Verify the correctness of the final sorted output and interpret the efficiency of Bubble Sort on already sorted data.

Q25. Linear Search – First Element Detection

Given the unsorted dataset [55, 21, 39, 47, 63, 70], apply Linear Search to find the element 55. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to find the target element. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target is found immediately.

Q26. Binary Search – Last Element Analysis

Given the sorted dataset [2, 6, 10, 14, 18, 22, 26, 30], apply Binary Search to find the element 30. Clearly interpret the problem and justify the use of Binary Search. Show each step involved in dividing the search space until the element is found. Count the total number of comparisons performed. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its efficiency when the target is at the last position.

Q27. Closest Pair – Real-World Tower Placement Scenario

A set of communication towers is represented by coordinates $\{(2,8), (5,6), (9,4), (4,7), (8,3)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest towers. Clearly interpret the problem and justify the selected method. Compute the distance between all tower pairs and show the calculations. Identify the closest pair and count the number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its practical importance in communication network planning.

Q28. Convex Hull – Outer Boundary of Survey Points

A land survey provides points $\{(0,1), (2,4), (5,5), (6,2), (4,0), (2,1)\}$. Apply the Brute Force Convex Hull algorithm to determine the outer boundary. Clearly interpret the problem and explain the need for convex hull computation. Check all possible line pairs and identify the valid hull edges. Show the logical steps used to exclude interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance in land mapping applications.

Q29. Sorting Techniques – Selection Sort vs Bubble Sort

Analyze the performance of Selection Sort and Bubble Sort on the dataset [28, 16, 35, 11, 19]. Clearly interpret the problem and explain the working principles of both algorithms. Apply both methods step-by-step and show intermediate results after each pass. Count the total number of comparisons and swaps in each case. Analyze their time complexities and compare their efficiency. Interpret the results and justify which method is preferable for this dataset.

Q30. Searching Techniques – Linear Search vs Binary Search

Given the dataset [7, 14, 21, 28, 35, 42, 49], analyze the performance of Linear Search and Binary Search to find the element 35. Clearly interpret the problem and explain the working of both searching methods. Show the step-by-step process for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency based on the dataset condition. Interpret the results and justify which method is preferable for practical use.

Q31. Selection Sort – Descending and Repeated Values

Apply the Brute Force technique using Selection Sort on the dataset [44, 31, 44, 18, 9, 18]. Clearly interpret the problem and justify why Selection Sort is appropriate for this comparison-based sorting task. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on repeated values.

Q32. Bubble Sort – Mixed Integer Dataset

Apply the Brute Force technique using Bubble Sort on the dataset [27, -5, 13, 0, 42, -8]. Clearly interpret the problem and explain why Bubble Sort can be applied to datasets containing both positive and negative numbers. Perform the sorting process pass by pass and show the intermediate results after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on mixed integers.

Q33. Linear Search – Duplicate Target Elements

Given the unsorted dataset [19, 42, 11, 42, 56, 73, 42], apply Linear Search to find the first occurrence of the element 42. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required until the first match is found. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when duplicate target values are present.

Q34. Binary Search – Even Number Sorted Dataset

Given the sorted dataset [4, 8, 12, 16, 20, 24, 28, 32], apply Binary Search to find the element 20. Clearly interpret the problem and justify why Binary Search is suitable for this ordered dataset. Show each step involved in dividing the search interval and identifying the middle element. Count the number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search on evenly spaced values.

Q35. Closest Pair – Dense Cluster of Points

Given the set of points {(2,2), (3,3), (3,4), (8,9), (10,11), (2,5)}, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the choice of the brute force method. Compute the distance between every possible pair and show the calculations clearly. Identify the closest pair and count the total number of comparisons made. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its relevance in cluster-based spatial analysis.

Q36. Convex Hull – Rectangular Boundary with Interior Points

Apply the Brute Force Convex Hull algorithm on the set {(0,0), (0,5), (5,5), (5,0), (2,2), (3,3), (1,4)}. Clearly interpret the problem and explain the concept of determining the outer boundary of a point set. Check all possible line pairs and identify which edges form the convex hull. Show the logical steps used to exclude interior points from the boundary. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q37. Selection Sort – Small Dataset with Minimum at End

Apply the Brute Force technique using Selection Sort on the dataset [21, 18, 14, 30, 6]. Clearly interpret the problem and justify the use of Selection Sort for identifying the minimum element in each pass. Perform the sorting step-by-step and show the array after each pass. Count the total number of comparisons and

swaps made. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q38. Bubble Sort – Alternating High and Low Values

Apply the Brute Force technique using Bubble Sort on the dataset [50, 10, 40, 20, 30, 5]. Clearly interpret the problem and explain how Bubble Sort repeatedly exchanges adjacent out-of-order elements. Perform the sorting process pass by pass and show all intermediate arrays. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the final sorted output and interpret the efficiency of the algorithm on alternating values.

Q39. Linear Search – Single Match in Middle Position

Given the unsorted dataset [14, 28, 35, 47, 59, 63, 71], apply Linear Search to find the element 47. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons needed before the target is found. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target lies near the middle of the list.

Q40. Binary Search – Search in Large Ordered List

Given the sorted dataset [5, 10, 15, 20, 25, 30, 35, 40, 45], apply Binary Search to find the element 40. Clearly interpret the problem and justify why Binary Search is appropriate for this dataset. Show the step-by-step reduction of the search interval. Count the number of comparisons required to reach the target. Analyze the time complexity of the method. Verify the correctness of the result and interpret its performance on larger ordered lists.

Q41. Closest Pair – Transportation Hub Coordinates

A set of transport hubs is represented by coordinates {(1,7), (4,9), (6,8), (3,5), (8,2), (2,6)}. Apply the Closest Pair Brute Force algorithm to identify the nearest hubs. Clearly interpret the problem and justify the chosen method. Compute the distance between all possible pairs and show the calculations clearly. Identify the closest pair and count the number of pairwise comparisons performed. Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its importance in transportation network planning.

Q42. Convex Hull – Pentagon with Inner Point

Apply the Brute Force Convex Hull algorithm on the set {(1,1), (3,5), (6,4), (7,2), (4,0), (4,2)}. Clearly interpret the problem and explain the concept of convex hull formation. Check all possible line pairs and determine which points form the outer boundary while excluding the interior point. Show the logical process used to validate the hull edges. Count the total comparisons performed. Analyze the time

complexity and interpret the usefulness of the result in geometric boundary detection.

Q43. Selection Sort – Dataset with Zero and Repetitions

Apply the Brute Force technique using Selection Sort on the dataset [16, 0, 9, 16, 4, 0]. Clearly interpret the problem and justify the use of Selection Sort for this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the effect of repeated zero values on the sorting process.

Q44. Bubble Sort – Nearly Reverse Ordered Data

Apply the Brute Force technique using Bubble Sort on the dataset [60, 50, 40, 30, 20, 25]. Clearly interpret the problem and explain why Bubble Sort is suitable for observing adjacent exchanges. Perform the sorting process step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on nearly reverse ordered data.

Q45. Linear Search – Negative Number Detection

Given the unsorted dataset [12, -7, 25, 39, -14, 48, 56], apply Linear Search to find the element -14. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required before locating the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when searching for negative values in an unsorted list.

Q46. Binary Search – Unsuccessful Search in Sorted List

Given the sorted dataset [11, 22, 33, 44, 55, 66, 77], apply Binary Search to find the element 50. Clearly interpret the problem and justify why Binary Search is used. Show all the steps involved in reducing the search interval until the search becomes unsuccessful. Count the number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search in unsuccessful cases.

Q47. Closest Pair – Points on a Grid

Given the set of points $\{(0,1), (1,2), (2,1), (3,3), (4,4), (2,2)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the chosen approach. Compute the distances between every possible pair and show the calculations. Identify the closest pair and count the number of comparisons performed. Analyze the time complexity of the brute force method. Verify the correctness of the result and interpret its significance in grid-based geometry problems.

Q48. Convex Hull – Wide Boundary Distribution

Apply the Brute Force Convex Hull algorithm on the set $\{(0,2), (2,6), (5,7), (8,5), (9,1), (4,0), (3,3)\}$. Clearly interpret the problem and explain the role of convex hull in identifying extreme points. Check all possible line pairs and identify the valid hull edges. Show the logical reasoning used to separate boundary points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q49. Sorting Techniques – Comparative Pass Analysis

Analyze the performance of Selection Sort and Bubble Sort on the dataset [33, 12, 27, 8, 19, 41]. Clearly interpret the problem and explain the working principle of both algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational behavior. Interpret the results and justify which method is more suitable for this dataset.

Q50. Searching Techniques – Ordered Data Efficiency Comparison

Given the sorted dataset [9, 18, 27, 36, 45, 54, 63, 72], analyze the performance of Linear Search and Binary Search to find the element 54. Clearly interpret the problem and explain the working process of both searching methods. Show the step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both methods. Compare their efficiency for ordered data. Interpret the results and justify which method is preferable in practical applications.

Q51. Selection Sort – Random Integer Arrangement

Apply the Brute Force technique using Selection Sort on the dataset [38, 12, 47, 5, 29, 18]. Clearly interpret the problem and justify why Selection Sort is suitable for this unsorted dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q52. Bubble Sort – Repeated Adjacent Disorder

Apply the Brute Force technique using Bubble Sort on the dataset [14, 35, 21, 42, 28, 7]. Clearly interpret the problem and explain how Bubble Sort resolves adjacent inversions through repeated passes. Perform the sorting process step-by-step and display the intermediate array after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on this dataset.

Q53. Linear Search – Target in Second Half

Given the unsorted dataset [61, 24, 13, 47, 32, 89, 53], apply Linear Search to find the element 89. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears near the end of the dataset.

Q54. Binary Search – Odd Length Sorted List

Given the sorted dataset [3, 9, 15, 21, 27, 33, 39], apply Binary Search to find the element 33. Clearly interpret the problem and justify why Binary Search is suitable for this ordered list. Show each step involved in selecting the middle element and reducing the search interval. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its efficiency.

Q55. Closest Pair – Academic Building Coordinates

A campus map gives building coordinates as {(1,4), (3,7), (6,8), (2,5), (9,2), (4,6)}. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of buildings. Clearly interpret the problem and justify the use of the brute force approach. Compute distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in campus planning.

Q56. Convex Hull – Outer Boundary of Facility Points

Apply the Brute Force Convex Hull algorithm on the set {(0,0), (2,4), (5,6), (8,3), (6,0), (3,2), (4,3)}. Clearly interpret the problem and explain the concept of identifying the outermost boundary. Check all possible line pairs and determine which edges belong to the convex hull. Show the logical reasoning used to exclude internal points. Count the total comparisons performed. Analyze the time complexity and interpret the practical importance of the result.

Q57. Selection Sort – Minimum Element Reappearing Later

Apply the Brute Force technique using Selection Sort on the dataset [26, 11, 34, 11, 52, 19]. Clearly interpret the problem and justify why Selection Sort is appropriate for repeated minimum candidates. Perform the sorting step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on duplicate minimum elements.

Q58. Bubble Sort – Smallest Element at End

Apply the Brute Force technique using Bubble Sort on the dataset [45, 38, 29, 17, 50, 6]. Clearly interpret the problem and explain how Bubble Sort moves the smallest misplaced element through repeated adjacent swaps. Perform the sorting pass by pass and show all intermediate arrays. Count the number of comparisons and swaps performed. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q59. Linear Search – Search in Mixed Dataset

Given the unsorted dataset [8, 19, 27, 44, 12, 31, 50], apply Linear Search to find the element 12. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons made until the element is found. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search for moderate-sized unsorted lists.

Q60. Binary Search – Lower Half Target Search

Given the sorted dataset [10, 20, 30, 40, 50, 60, 70, 80], apply Binary Search to find the element 20. Clearly interpret the problem and justify the use of Binary Search for this ordered dataset. Show the step-by-step reduction of the search interval. Count the number of comparisons required to locate the target. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm.

Q61. Closest Pair – Warehouse Location Analysis

A logistics company has warehouses located at $\{(2,1), (5,4), (6,5), (10,9), (3,2), (8,7)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest warehouses. Clearly interpret the problem and justify the chosen method. Compute the distances between all pairs and show the calculations clearly. Identify the closest pair and count the number of pairwise comparisons performed. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its use in logistics optimization.

Q62. Convex Hull – Boundary Detection with Central Points

Apply the Brute Force Convex Hull algorithm on the set $\{(1,0), (2,5), (6,6), (8,2), (5,0), (4,3), (3,2)\}$. Clearly interpret the problem and explain the concept of convex hull. Check all possible line pairs and identify the valid hull edges that form the external boundary. Show the reasoning used to reject interior points. Count the total comparisons performed. Analyze the time complexity and interpret the usefulness of the hull in geometric applications.

Q63. Selection Sort – Dataset with Large and Small Values

Apply the Brute Force technique using Selection Sort on the dataset [99, 4, 76, 15, 2, 61]. Clearly interpret the problem and justify why Selection Sort can be applied effectively to this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the number of comparisons and swaps

made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q64. Bubble Sort – Duplicate and Ordered Blocks

Apply the Brute Force technique using Bubble Sort on the dataset [11, 11, 25, 18, 18, 7]. Clearly interpret the problem and explain the effect of adjacent comparisons on duplicate-valued elements. Perform the sorting process pass by pass and show the intermediate arrays. Count the total number of comparisons and swaps made. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on partially grouped values.

Q65. Linear Search – Unsuccessful Search in Unsorted Data

Given the unsorted dataset [16, 28, 39, 47, 58, 69, 74], apply Linear Search to find the element 100. Clearly interpret the problem and explain the sequential search process in detail. Count the total number of comparisons made before concluding that the element is absent. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search in unsuccessful cases.

Q66. Binary Search – Central Region Search

Given the sorted dataset [7, 14, 21, 28, 35, 42, 49, 56, 63], apply Binary Search to find the element 42. Clearly interpret the problem and justify why Binary Search is appropriate for this dataset. Show the sequence of middle elements examined and the reduction of the search space. Count the number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm.

Q67. Closest Pair – Sensor Node Placement

A monitoring system places sensor nodes at $\{(0,3), (2,4), (5,7), (6,8), (1,2), (9,9)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest sensor nodes. Clearly interpret the problem and justify the use of the brute force approach. Compute all pairwise distances and show the calculations clearly. Identify the closest pair and count the total number of comparisons. Analyze the time complexity of the method. Verify the correctness of the result and interpret its relevance in sensor network design.

Q68. Convex Hull – Irregular Outer Shape Analysis

Apply the Brute Force Convex Hull algorithm on the set $\{(0,1), (1,5), (4,7), (7,6), (9,2), (6,0), (3,1), (4,3)\}$. Clearly interpret the problem and explain how the convex hull captures the extreme boundary points. Check all possible line pairs and identify the valid hull edges. Show the logical procedure used to exclude inner points from the hull. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q69. Sorting Techniques – Detailed Comparison of Two Methods

Analyze the performance of Selection Sort and Bubble Sort on the dataset [48, 22, 35, 10, 27, 16]. Clearly interpret the problem and explain the working principles of both algorithms. Apply both sorting methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational differences. Interpret the results and justify which method is preferable for this dataset.

Q70. Searching Techniques – Practical Efficiency Study

Given the sorted dataset [6, 12, 18, 24, 30, 36, 42, 48], analyze the performance of Linear Search and Binary Search to find the element 36. Clearly interpret the problem and explain the working process of both searching algorithms. Show the step-by-step procedure for each method and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for sorted data. Interpret the results and justify which method is preferable for practical use.

Q71. Selection Sort – Alternating High and Low Values

Apply the Brute Force technique using Selection Sort on the dataset [52, 11, 47, 8, 39, 6]. Clearly interpret the problem and justify why Selection Sort is suitable for this unsorted dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on alternating values.

Q72. Bubble Sort – Mixed Ordered Segments

Apply the Brute Force technique using Bubble Sort on the dataset [9, 18, 27, 14, 22, 5]. Clearly interpret the problem and explain how Bubble Sort compares adjacent elements to resolve local disorder. Perform the sorting pass by pass and display the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on partially ordered segments.

Q73. Linear Search – Search for Repeated Nonconsecutive Target

Given the unsorted dataset [31, 14, 29, 14, 52, 67, 14], apply Linear Search to find the first occurrence of the element 14. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when repeated target values are scattered across the list.

Q74. Binary Search – Target Near Upper Middle Region

Given the sorted dataset [2, 6, 10, 14, 18, 22, 26, 30, 34], apply Binary Search to find the element 22. Clearly interpret the problem and justify why Binary Search is appropriate for this ordered dataset. Show each step involved in finding the middle element and reducing the search interval. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its efficiency.

Q75. Closest Pair – Hospital Location Planning

A city map provides hospital coordinates as {(1,2), (3,5), (6,7), (2,3), (8,1), (5,6)}. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of hospitals. Clearly interpret the problem and justify the use of the brute force approach. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in healthcare access planning.

Q76. Convex Hull – Boundary Detection in Urban Layout

Apply the Brute Force Convex Hull algorithm on the set {(0,0), (2,5), (5,7), (8,4), (7,0), (4,2), (3,3)}. Clearly interpret the problem and explain the concept of identifying the convex boundary. Check all possible line pairs and determine which edges belong to the hull. Show the reasoning used to distinguish boundary points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in urban mapping.

Q77. Selection Sort – Dataset with Two Smallest Equal Values

Apply the Brute Force technique using Selection Sort on the dataset [24, 7, 31, 7, 45, 12]. Clearly interpret the problem and justify why Selection Sort is suitable for identifying the minimum element in each pass. Perform the sorting step-by-step and show the intermediate arrays after each pass. Count the number of comparisons and swaps made during execution. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on equal minimum values.

Q78. Bubble Sort – Large Element Drift Analysis

Apply the Brute Force technique using Bubble Sort on the dataset [13, 41, 19, 36, 8, 50]. Clearly interpret the problem and explain how Bubble Sort gradually moves larger elements toward their correct positions. Perform the sorting pass by pass and show all intermediate arrays. Count the total number of comparisons and swaps performed. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the algorithm's operational efficiency.

Q79. Linear Search – Target Immediately After First Element

Given the unsorted dataset [60, 25, 33, 41, 52, 77, 89], apply Linear Search to find the element 25. Clearly interpret the problem and explain the search process step-

by-step. Count the number of comparisons made before finding the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears near the beginning of the list.

Q80. Binary Search – Unsuccessful Search Above Maximum Value

Given the sorted dataset [5, 15, 25, 35, 45, 55, 65, 75], apply Binary Search to find the element 90. Clearly interpret the problem and justify the method selection. Show all the steps involved in reducing the search space until the search becomes unsuccessful. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search in unsuccessful search scenarios.

Q81. Closest Pair – Distribution of Delivery Centers

A distribution company has delivery centers at {(2,4), (4,7), (7,8), (3,5), (9,2), (6,6)}. Apply the Closest Pair Brute Force algorithm to identify the nearest centers. Clearly interpret the problem and justify the chosen method. Compute the distances between all possible pairs and show the calculations step-by-step. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its usefulness in route optimization.

Q82. Convex Hull – Hexagonal Spread with Inner Points

Apply the Brute Force Convex Hull algorithm on the set {(1,1), (3,6), (6,7), (9,4), (8,1), (4,0), (5,3), (4,2)}. Clearly interpret the problem and explain how the convex hull captures the extreme boundary points. Check all possible line pairs and identify the edges that form the hull. Show the logical reasoning used to reject interior points. Count the total comparisons performed. Analyze the time complexity and interpret the result in computational geometry.

Q83. Selection Sort – High Variance Dataset

Apply the Brute Force technique using Selection Sort on the dataset [70, 3, 58, 21, 94, 11]. Clearly interpret the problem and justify why Selection Sort is appropriate for this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on highly varied values.

Q84. Bubble Sort – Repeated Local Inversions

Apply the Brute Force technique using Bubble Sort on the dataset [26, 17, 35, 29, 48, 12]. Clearly interpret the problem and explain how Bubble Sort resolves multiple local inversions through repeated adjacent comparisons. Perform the sorting process pass by pass and show the intermediate arrays. Count the total number of comparisons and swaps made during execution. Analyze the time

complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on this dataset.

Q85. Linear Search – Target in Final Two Positions

Given the unsorted dataset [18, 24, 37, 49, 55, 63, 71], apply Linear Search to find the element 63. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target is located near the end of the list.

Q86. Binary Search – Exact Center Value Search

Given the sorted dataset [8, 16, 24, 32, 40, 48, 56], apply Binary Search to find the element 32. Clearly interpret the problem and justify why Binary Search is ideal for this ordered list. Show the step-by-step reduction of the search interval. Count the total number of comparisons made before locating the target. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of Binary Search when the target is exactly at the center.

Q87. Closest Pair – Monitoring Station Coordinates

A monitoring system uses station coordinates $\{(0,2), (2,4), (5,5), (6,6), (1,3), (8,1)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of stations. Clearly interpret the problem and justify the use of the brute force technique. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the number of pairwise comparisons made. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its relevance in monitoring network design.

Q88. Convex Hull – Outer Shape of Surveyed Coordinates

Apply the Brute Force Convex Hull algorithm on the set $\{(0,1), (2,6), (5,8), (8,6), (9,2), (6,0), (3,1), (4,4)\}$. Clearly interpret the problem and explain the purpose of finding the convex hull. Check all possible line pairs and determine which edges form the external boundary. Show the logical steps used to separate hull points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in land survey analysis.

Q89. Sorting Techniques – Comparison on Moderate Sized Dataset

Analyze the performance of Selection Sort and Bubble Sort on the dataset [54, 23, 41, 12, 36, 18]. Clearly interpret the problem and explain the working principles of both sorting algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational efficiency. Interpret the results and justify which method is more suitable for this dataset.

Q90. Searching Techniques – Efficiency Comparison on Sorted Dataset

Given the sorted dataset [4, 8, 12, 16, 20, 24, 28, 32, 36], analyze the performance of Linear Search and Binary Search to find the element 28. Clearly interpret the problem and explain the working of both searching methods. Show the step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for sorted data. Interpret the results and justify which method is preferable in practical applications.

Q91. Selection Sort – Dataset with Scattered Small Elements

Apply the Brute Force technique using Selection Sort on the dataset [43, 9, 37, 14, 28, 5]. Clearly interpret the problem and justify why Selection Sort is suitable for identifying the minimum element during each pass. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on this dataset.

Q92. Bubble Sort – Progressive Exchange Analysis

Apply the Brute Force technique using Bubble Sort on the dataset [34, 12, 29, 45, 18, 7]. Clearly interpret the problem and explain how Bubble Sort repeatedly compares adjacent elements and exchanges them when necessary. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the operational efficiency of the algorithm.

Q93. Linear Search – Search for Middle-Late Position Target

Given the unsorted dataset [22, 31, 47, 58, 16, 39, 64], apply Linear Search to find the element 39. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears in the latter half of the dataset.

Q94. Binary Search – Search in Uniformly Spaced Dataset

Given the sorted dataset [5, 10, 15, 20, 25, 30, 35, 40], apply Binary Search to find the element 25. Clearly interpret the problem and justify why Binary Search is appropriate for this ordered dataset. Show the step-by-step reduction of the search interval by checking the middle element. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of the algorithm.

Q95. Closest Pair – Emergency Service Location Analysis

A city has emergency service locations at $\{(1,3), (4,5), (7,8), (2,4), (9,1), (5,6)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest service locations. Clearly interpret the problem and justify the use of the brute force method. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its significance in emergency response planning.

Q96. Convex Hull – Boundary Points in Open Field Mapping

Apply the Brute Force Convex Hull algorithm on the set $\{(0,0), (2,6), (5,8), (8,5), (7,1), (4,2), (3,4)\}$. Clearly interpret the problem and explain the concept of finding the convex hull of a set of points. Check all possible line pairs and identify the edges that form the outer boundary. Show the logical reasoning used to exclude inner points from the hull. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in field mapping.

Q97. Selection Sort – Duplicate Middle Elements

Apply the Brute Force technique using Selection Sort on the dataset [18, 27, 18, 42, 9, 33]. Clearly interpret the problem and justify why Selection Sort can be applied effectively to this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm when duplicate elements are present.

Q98. Bubble Sort – Unordered Tail Segment

Apply the Brute Force technique using Bubble Sort on the dataset [10, 20, 30, 15, 25, 5]. Clearly interpret the problem and explain how Bubble Sort resolves disorder in the latter part of the dataset through adjacent comparisons. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q99. Linear Search – Repeated Value Search

Given the unsorted dataset [13, 26, 39, 26, 52, 65, 78], apply Linear Search to find the first occurrence of the element 26. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when repeated values are present.

Q100. Binary Search – Unsuccessful Search Below Minimum Value

Given the sorted dataset [12, 24, 36, 48, 60, 72, 84], apply Binary Search to find the element 6. Clearly interpret the problem and justify the method selection. Show all the steps involved in repeatedly reducing the search space until the search becomes unsuccessful. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search in this unsuccessful search case.

Q101. Closest Pair – Retail Outlet Distribution

A retail chain has outlets located at $\{(2,2), (4,4), (7,7), (3,3), (8,1), (6,5)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest outlets. Clearly interpret the problem and justify the use of the brute force technique. Compute distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its importance in retail distribution planning.

Q102. Convex Hull – Extreme Point Identification in Regional Layout

Apply the Brute Force Convex Hull algorithm on the set $\{(1,0), (2,5), (5,7), (8,6), (9,2), (6,0), (4,3), (3,2)\}$. Clearly interpret the problem and explain how the convex hull identifies the extreme points of a geometric layout. Check all possible line pairs and determine the valid hull edges. Show the logical procedure used to separate boundary points from interior points. Count the total comparisons performed. Analyze the time complexity and interpret the practical value of the result in regional layout analysis.

Q103. Selection Sort – High-Low Mixed Arrangement

Apply the Brute Force technique using Selection Sort on the dataset [64, 11, 58, 23, 7, 49]. Clearly interpret the problem and justify why Selection Sort is suitable for this comparison-based sorting task. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q104. Bubble Sort – Duplicate Ends Analysis

Apply the Brute Force technique using Bubble Sort on the dataset [25, 14, 31, 19, 25, 8]. Clearly interpret the problem and explain how Bubble Sort handles duplicate elements during repeated adjacent comparisons. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on datasets containing repeated end values.

Q105. Linear Search – Search for Minimum Value

Given the unsorted dataset [44, 18, 29, 7, 53, 61, 72], apply Linear Search to find the element 7. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons needed to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when searching for the minimum value in an unsorted list.

Q106. Binary Search – Upper Region Successful Search

Given the sorted dataset [9, 18, 27, 36, 45, 54, 63, 72], apply Binary Search to find the element 63. Clearly interpret the problem and justify the use of Binary Search for this dataset. Show the sequence of middle elements examined and the reduction of the search interval. Count the total number of comparisons made before locating the target. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of the algorithm.

Q107. Closest Pair – Satellite Station Coordinates

A satellite communication network uses station coordinates $\{(0,1), (2,3), (5,6), (6,7), (1,2), (9,4)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of stations. Clearly interpret the problem and justify the chosen method. Compute the distances between all possible pairs and show the calculations step-by-step. Identify the closest pair and count the total number of pairwise comparisons. Analyze the time complexity of the approach. Verify the correctness of the result and interpret its significance in communication network design.

Q108. Convex Hull – Geometric Boundary with Internal Cluster

Apply the Brute Force Convex Hull algorithm on the set $\{(0,2), (2,7), (5,9), (8,7), (10,3), (7,0), (3,1), (5,4)\}$. Clearly interpret the problem and explain the role of convex hull in identifying the external geometric boundary. Check all possible line pairs and identify the edges that form the hull. Show the logical reasoning used to reject the internal clustered points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q109. Sorting Techniques – Efficiency Comparison on Diverse Values

Analyze the performance of Selection Sort and Bubble Sort on the dataset [62, 17, 45, 9, 28, 33]. Clearly interpret the problem and explain the working principles of both sorting algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational behavior. Interpret the results and justify which method is preferable for this dataset.

Q110. Searching Techniques – Comparison on Ordered List

Given the sorted dataset [7, 14, 21, 28, 35, 42, 49, 56], analyze the performance of Linear Search and Binary Search to find the element 49. Clearly interpret the problem and explain the working process of both searching methods. Show the

step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for ordered data. Interpret the results and justify which method is preferable in practical applications.

Q111. Selection Sort – Widely Distributed Values

Apply the Brute Force technique using Selection Sort on the dataset [72, 14, 59, 6, 38, 21]. Clearly interpret the problem and justify why Selection Sort is suitable for repeatedly selecting the minimum element from the unsorted portion. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on this dataset.

Q112. Bubble Sort – Late Disorder Resolution

Apply the Brute Force technique using Bubble Sort on the dataset [11, 24, 37, 18, 29, 7]. Clearly interpret the problem and explain how Bubble Sort resolves disorder through repeated adjacent comparisons and exchanges. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the operational efficiency of the algorithm.

Q113. Linear Search – Target in the Final Position

Given the unsorted dataset [27, 35, 41, 53, 62, 74, 88], apply Linear Search to find the element 88. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears at the final position.

Q114. Binary Search – Upper Half Ordered Search

Given the sorted dataset [4, 9, 14, 19, 24, 29, 34, 39, 44], apply Binary Search to find the element 34. Clearly interpret the problem and justify why Binary Search is appropriate for this ordered dataset. Show the step-by-step reduction of the search interval by evaluating the middle element at each stage. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its efficiency.

Q115. Closest Pair – Weather Station Proximity Analysis

A region contains weather stations at coordinates $\{(1,1), (3,4), (6,6), (2,2), (8,1), (5,5)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest pair of stations. Clearly interpret the problem and justify the use of the brute force approach. Compute distances between all possible pairs and show the

calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in station network planning.

Q116. Convex Hull – Boundary Formation in Resource Mapping

Apply the Brute Force Convex Hull algorithm on the set $\{(0,1), (2,6), (5,8), (8,6), (9,2), (6,0), (3,1), (4,4)\}$. Clearly interpret the problem and explain the concept of determining the convex boundary of a set of points. Check all possible line pairs and identify which edges form the hull. Show the logical reasoning used to exclude the inner points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in resource mapping.

Q117. Selection Sort – Duplicate Large Values Analysis

Apply the Brute Force technique using Selection Sort on the dataset [31, 48, 12, 48, 7, 25]. Clearly interpret the problem and justify why Selection Sort is suitable for this dataset containing repeated large values. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on duplicated elements.

Q118. Bubble Sort – Mixed Small and Large Elements

Apply the Brute Force technique using Bubble Sort on the dataset [42, 9, 33, 15, 27, 3]. Clearly interpret the problem and explain how Bubble Sort gradually places larger elements toward the end of the list. Perform the sorting process step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on mixed values.

Q119. Linear Search – First Repeated Target Match

Given the unsorted dataset [16, 28, 16, 39, 52, 64, 16], apply Linear Search to find the first occurrence of the element 16. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears repeatedly in the list.

Q120. Binary Search – Search for Lower Midrange Element

Given the sorted dataset [6, 12, 18, 24, 30, 36, 42, 48], apply Binary Search to find the element 24. Clearly interpret the problem and justify the method selection. Show all the steps involved in repeatedly dividing the search interval until the target is found. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search on sorted numerical data.

Q121. Closest Pair – Public Service Point Distribution

A municipality has service points at $\{(2,3), (4,6), (7,7), (3,4), (9,2), (6,5)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest service points. Clearly interpret the problem and justify the chosen brute force approach. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of pairwise comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its usefulness in public service planning.

Q122. Convex Hull – Point Set with Internal Core

Apply the Brute Force Convex Hull algorithm on the set $\{(1,1), (3,7), (6,8), (9,5), (8,1), (4,0), (5,3), (4,2)\}$. Clearly interpret the problem and explain how the convex hull captures only the extreme outer boundary points. Check all possible line pairs and identify the valid hull edges. Show the logical steps used to reject the internal core points. Count the total comparisons performed. Analyze the time complexity and interpret the result in terms of geometric boundary detection.

Q123. Selection Sort – Unordered Values with Repeated Minimum

Apply the Brute Force technique using Selection Sort on the dataset [22, 5, 31, 14, 5, 27]. Clearly interpret the problem and justify why Selection Sort is suitable for identifying the smallest element in each pass. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the algorithm's behavior on repeated minimum values.

Q124. Bubble Sort – Reverse Tendency with Central Disorder

Apply the Brute Force technique using Bubble Sort on the dataset [50, 41, 32, 23, 35, 8]. Clearly interpret the problem and explain how Bubble Sort uses adjacent exchanges to eliminate inversions. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on this partially reverse-ordered dataset.

Q125. Linear Search – Search for Smallest Existing Element

Given the unsorted dataset [45, 19, 33, 7, 52, 68, 81], apply Linear Search to find the element 7. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when searching for a smaller value positioned in the middle portion of the list.

Q126. Binary Search – Search for Highest Existing Element

Given the sorted dataset [8, 16, 24, 32, 40, 48, 56, 64], apply Binary Search to find the element 64. Clearly interpret the problem and justify why Binary Search is suitable for this ordered dataset. Show the step-by-step reduction of the search interval until the target is found. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of the algorithm when the target is the largest element.

Q127. Closest Pair – Mobile Tower Layout Analysis

A network provider has mobile towers at coordinates {(0,2), (2,5), (5,7), (6,8), (1,3), (9,1)}. Apply the Closest Pair Brute Force algorithm to identify the nearest towers. Clearly interpret the problem and justify the use of the brute force method. Compute distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in tower layout optimization.

Q128. Convex Hull – Survey Boundary with Interior Cluster

Apply the Brute Force Convex Hull algorithm on the set {(0,0), (2,7), (5,9), (8,7), (10,3), (7,0), (3,1), (5,4)}. Clearly interpret the problem and explain the role of convex hull in determining the external survey boundary. Check all possible line pairs and identify the valid hull edges. Show the logical reasoning used to reject the interior clustered points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in surveying applications.

Q129. Sorting Techniques – Comparative Operational Study

Analyze the performance of Selection Sort and Bubble Sort on the dataset [57, 21, 43, 12, 36, 18]. Clearly interpret the problem and explain the working principles of both sorting algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational characteristics. Interpret the results and justify which method is more suitable for this dataset.

Q130. Searching Techniques – Ordered Dataset Performance Comparison

Given the sorted dataset [5, 10, 15, 20, 25, 30, 35, 40, 45], analyze the performance of Linear Search and Binary Search to find the element 40. Clearly interpret the problem and explain the working of both searching methods. Show the step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for ordered data. Interpret the results and justify which method is preferable in practical applications.

Q131. Selection Sort – Dataset with Alternating Extremes

Apply the Brute Force technique using Selection Sort on the dataset [91, 3, 78, 5, 64, 7]. Clearly interpret the problem and justify why Selection Sort is suitable for repeatedly selecting the minimum element from the unsorted portion. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on this dataset.

Q132. Bubble Sort – High Disorder Dataset

Apply the Brute Force technique using Bubble Sort on the dataset [29, 11, 47, 6, 38, 14]. Clearly interpret the problem and explain how Bubble Sort resolves disorder through repeated adjacent comparisons. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in different cases. Verify the correctness of the final sorted output and interpret the operational efficiency of the algorithm.

Q133. Linear Search – Target Appearing Multiple Times

Given the unsorted dataset [23, 45, 23, 67, 89, 23, 12], apply Linear Search to find the first occurrence of the element 23. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears multiple times.

Q134. Binary Search – Target Near Lower Region

Given the sorted dataset [6, 12, 18, 24, 30, 36, 42, 48], apply Binary Search to find the element 12. Clearly interpret the problem and justify why Binary Search is appropriate for this ordered dataset. Show the step-by-step reduction of the search interval by evaluating the middle element at each stage. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its efficiency.

Q135. Closest Pair – Industrial Unit Location Analysis

An industrial layout has units at coordinates $\{(1,2), (4,5), (7,7), (2,3), (9,2), (5,6)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest units. Clearly interpret the problem and justify the use of the brute force approach. Compute distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in industrial planning.

Q136. Convex Hull – Boundary Detection in Land Survey

Apply the Brute Force Convex Hull algorithm on the set $\{(0,1), (2,6), (5,9), (8,7), (9,3), (6,0), (3,1), (4,5)\}$. Clearly interpret the problem and explain the concept of determining the convex boundary of a set of points. Check all possible line pairs and identify which edges form the hull. Show the logical reasoning used to exclude inner points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in surveying applications.

Q137. Selection Sort – Dataset with Repeated Largest Values

Apply the Brute Force technique using Selection Sort on the dataset [14, 52, 31, 52, 9, 27]. Clearly interpret the problem and justify why Selection Sort is suitable for this dataset containing repeated maximum values. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the behavior of the algorithm on duplicated maximum elements.

Q138. Bubble Sort – Mixed Increasing and Decreasing Blocks

Apply the Brute Force technique using Bubble Sort on the dataset [8, 16, 24, 32, 20, 12]. Clearly interpret the problem and explain how Bubble Sort resolves disorder in partially sorted blocks. Perform the sorting process pass by pass and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in all cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on mixed sequences.

Q139. Linear Search – Target Near Middle-End Position

Given the unsorted dataset [11, 22, 33, 44, 55, 66, 77], apply Linear Search to find the element 66. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target is positioned near the end.

Q140. Binary Search – Search for Central-Upper Element

Given the sorted dataset [5, 10, 15, 20, 25, 30, 35, 40, 45], apply Binary Search to find the element 35. Clearly interpret the problem and justify why Binary Search is suitable for this ordered dataset. Show the sequence of middle elements examined and the reduction of the search interval. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm.

Q141. Closest Pair – Smart City Sensor Distribution

A smart city deploys sensors at $\{(2,3), (4,6), (7,8), (3,4), (9,1), (6,5)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest sensors. Clearly interpret the problem and justify the use of the brute force technique. Compute the distances between all possible pairs and show the calculations clearly. Identify the

closest pair and count the total number of pairwise comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in sensor placement optimization.

Q142. Convex Hull – Boundary Detection with Irregular Points

Apply the Brute Force Convex Hull algorithm on the set $\{(0,2), (2,7), (5,10), (8,8), (10,4), (7,0), (3,1), (5,5)\}$. Clearly interpret the problem and explain how the convex hull identifies the extreme outer points. Check all possible line pairs and determine which edges form the hull. Show the logical reasoning used to reject the internal points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result in irregular boundary detection.

Q143. Selection Sort – Dataset with Closely Spaced Values

Apply the Brute Force technique using Selection Sort on the dataset [21, 19, 23, 18, 22, 20]. Clearly interpret the problem and justify why Selection Sort is suitable for this dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm on closely spaced values.

Q144. Bubble Sort – Repeated Small Element Movement

Apply the Brute Force technique using Bubble Sort on the dataset [30, 25, 20, 15, 10, 5]. Clearly interpret the problem and explain how Bubble Sort repeatedly moves the smallest element toward the beginning through successive passes. Perform the sorting process step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps performed. Analyze the time complexity in worst-case scenarios. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm.

Q145. Linear Search – Immediate Match Scenario

Given the unsorted dataset [99, 18, 27, 36, 45, 54, 63], apply Linear Search to find the element 99. Clearly interpret the problem and explain the search process step-by-step. Count the number of comparisons required to locate the target. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target is found immediately.

Q146. Binary Search – Unsuccessful Search Within Range Gap

Given the sorted dataset [10, 20, 30, 40, 50, 60, 70], apply Binary Search to find the element 35. Clearly interpret the problem and justify the method selection. Show all the steps involved in reducing the search interval until the element is determined to be absent. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the efficiency of Binary Search in unsuccessful cases.

Q147. Closest Pair – Network Node Placement

A communication network has nodes at $\{(1,3), (3,5), (6,7), (2,4), (8,2), (5,6)\}$. Apply the Closest Pair Brute Force algorithm to identify the nearest nodes. Clearly interpret the problem and justify the chosen approach. Compute the distances between all possible pairs and show the calculations clearly. Identify the closest pair and count the total number of comparisons performed. Analyze the time complexity of the method. Verify the correctness of the result and interpret its significance in network optimization.

Q148. Convex Hull – Outer Shape Identification in Data Points

Apply the Brute Force Convex Hull algorithm on the set $\{(0,0), (2,8), (5,10), (9,7), (10,3), (7,0), (3,2), (5,6)\}$. Clearly interpret the problem and explain how the convex hull determines the outer shape of a point set. Check all possible line pairs and identify the valid hull edges. Show the logical reasoning used to exclude internal points. Count the total comparisons performed. Analyze the time complexity and interpret the practical significance of the result.

Q149. Sorting Techniques – Comparative Efficiency Evaluation

Analyze the performance of Selection Sort and Bubble Sort on the dataset [46, 17, 39, 8, 28, 21]. Clearly interpret the problem and explain the working principles of both sorting algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational efficiency. Interpret the results and justify which method is more suitable for this dataset.

Q150. Searching Techniques – Final Comparative Analysis

Given the sorted dataset [6, 12, 18, 24, 30, 36, 42, 48, 54], analyze the performance of Linear Search and Binary Search to find the element 42. Clearly interpret the problem and explain the working of both searching methods. Show the step-by-step procedure for each algorithm and count the number of comparisons required. Analyze the time complexity of both approaches. Compare their efficiency for ordered data. Interpret the results and justify which method is preferable in practical applications.

RUBRICS FOR CALCULATION

Analytical Problem Solving					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretatio n of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretati on with minor omissions	Limited or unclear interpretation	No interpretati on
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